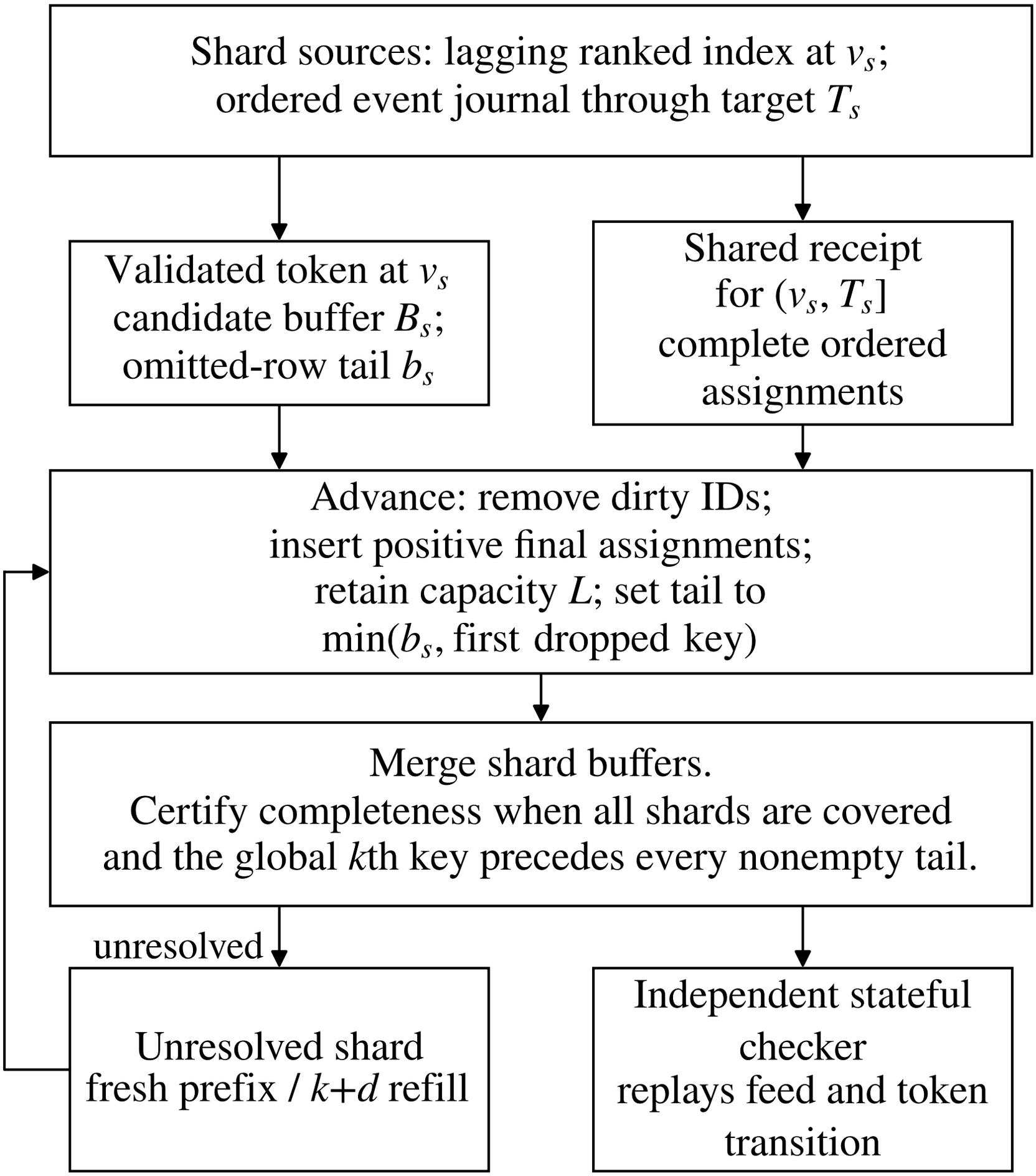
Shard sources: lagging ranked index at v_s;
ordered event journal through target T_s
Validated token at v_s
candidate buffer B_s;
omitted-row tail b_s
Shared receipt
for $(v_s, T_s]$
complete ordered
assignments
Advance: remove dirty IDs;
insert positive final assignments;
retain capacity L; set tail to
min(b_s, first dropped key)
Merge shard buffers.
Certify completeness when all shards are covered
and the global kth key precedes every nonempty tail.
unresolved
Unresolved shard
fresh prefix / k+d refill
Independent stateful
checker
replays feed and token
transition